■ Credit Wise – Loan Approval Prediction

Machine Learning Project README

Project Overview

Credit Wise is a Machine Learning project that predicts whether a loan application will be **approved or rejected** based on applicant financial, personal, employment, and credit-related information.

The project includes data preprocessing, exploratory data analysis (EDA), categorical encoding, feature scaling, machine learning model training, evaluation, and feature engineering.

■ Objective

The main objective of this project is to build a Machine Learning classification system that can predict:

• **Yes** → Loan Approved
• **No** → Loan Not Approved

The project also compares different classification algorithms using evaluation metrics such as:

• Precision
• Recall
• F1 Score
• Accuracy
• Confusion Matrix

■ Dataset

The dataset contains **1,000 loan applications** with **20 columns**.

Important Features

Feature	Description
Applicant_ID	Unique applicant identifier
Applicant_Income	Applicant's income
Coapplicant_Income	Co-applicant's income
Employment_Status	Employment type/status
Age	Applicant age
Marital_Status	Marital status
Dependents	Number of dependents
Credit_Score	Credit score
Existing_Loans	Existing loans
DTI_Ratio	Debt-to-income ratio
Savings	Applicant savings
Collateral_Value	Value of collateral
Loan_Amount	Requested loan amount
Loan_Term	Loan repayment term
Loan_Purpose	Purpose of the loan
Property_Area	Property area
Education_Level	Education level
Gender	Applicant gender

Feature	Description
Employer_Category	Employer category
Loan_Approved	Target variable

Target Variable

Loan_Approved

• **Yes** → Loan approved

• **No** → Loan rejected

■ Project Workflow

```
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Model Re-training & Evaluation
```

■ Data Preprocessing

1. Missing Value Handling

For numerical columns:

```
SimpleImputer(strategy="mean")
```

For categorical columns:

```
SimpleImputer(strategy="most_frequent")
```

This helps handle missing values before applying machine learning algorithms.

2. Removing Unnecessary Feature

Applicant_ID was removed because it is an identifier and does not provide useful predictive information.

```
df = df.drop("Applicant_ID", axis=1)
```

■ Exploratory Data Analysis

Several visualizations were created to understand the dataset.

Analysis performed:

• Loan approval class distribution

• Gender distribution

• Education level distribution

• Applicant income distribution

• Co-applicant income distribution
• Income outlier analysis
• Credit Score distribution
• DTI Ratio analysis
• Savings analysis
• Loan approval vs applicant income
• Loan approval vs credit score

Libraries used for visualization:
• Matplotlib
• Seaborn

■ Feature Encoding

Categorical features were converted into numerical values so that they could be used by Machine Learning models.

Label Encoding

Label Encoding was applied to:
• **Education_Level**
• **Loan_Approved**

One-Hot Encoding

One-Hot Encoding was applied to:
• **Employment_Status**
• **Marital_Status**
• **Loan_Purpose**
• **Property_Area**
• **Gender**
• **Employer_Category**

```
OneHotEncoder(
    drop="first",
    sparse_output=False,
    handle_unknown="ignore"
)
```

■ Correlation Analysis

A correlation matrix and heatmap were created to analyze relationships between numerical features.

```
corr_matrix = num_cols.corr()
```

A correlation heatmap was also generated using Seaborn.

The project additionally checks the correlation of features with the target variable:

```
num_cols.corr()["Loan_Approved"].sort_values(ascending=False)
```

✂■ Train-Test Split

The dataset was divided into training and testing data.

```
train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

Split
• **80%** → Training data

• **20%** → Testing data

■ Feature Scaling

StandardScaler was used to standardize the features.

```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Feature scaling is particularly useful for models such as Logistic Regression and KNN.

■ Machine Learning Models

1. Logistic Regression

Logistic Regression was trained as a binary classification model.

```
log_model = LogisticRegression()

log_model.fit(
    X_train_scaled,
    y_train
)
```

2. K-Nearest Neighbors (KNN)

KNN was implemented using:

```
KNeighborsClassifier(n_neighbors=7)
```

The model was trained using the scaled dataset.

3. Naive Bayes

Gaussian Naive Bayes was used for classification.

```
GaussianNB()
```

■ Model Evaluation

The models were evaluated using:

Precision

Measures how many predicted positive cases were actually positive.

Recall

Measures how many actual positive cases were correctly identified.

F1 Score

Provides a balance between precision and recall.

Accuracy

Measures the overall percentage of correct predictions.

Confusion Matrix

Shows:
• True Positive
• True Negative
• False Positive

• False Negative

Evaluation metrics used in the project:

```
precision_score()
recall_score()
f1_score()
accuracy_score()
confusion_matrix()
```

■■ Feature Engineering

Additional features were created to improve the representation of the data.

DTI Ratio Squared

```
df["DTI_Ratio_sq"] = df["DTI_Ratio"] ** 2
```

Credit Score Squared

```
df["Credit_Score_sq"] = df["Credit_Score"] ** 2
```

Applicant Income Log Transformation

```
df["Applicant_Income_log"] = np.log1p(df["Applicant_Income"])
```

The original **Credit_Score** and **DTI_Ratio** features were then removed from the final feature set in the feature-engineering stage.

■ Technologies Used

Programming Language
• Python

Libraries
• NumPy
• Pandas
• Matplotlib
• Seaborn
• Scikit-learn

Machine Learning
• Logistic Regression
• K-Nearest Neighbors
• Gaussian Naive Bayes

Data Processing
• SimpleImputer
• LabelEncoder
• OneHotEncoder
• StandardScaler
• Train-Test Split

■ Project Structure

```
Credit-Wise/
■
■■■ credit_wise.ipynb
■■■ loan_approval_data.csv
■■■ README.md
■■■ requirements.txt
```

Note: The notebook currently loads the dataset using **loan_approval_data.csv**. If your local dataset file has a different extension/name, rename it or update the path in the notebook accordingly.

■ How to Run the Project

1. Clone the repository

```
git clone <YOUR_GITHUB_REPOSITORY_URL>
```

2. Open the project folder

```
cd Credit-Wise
```

3. Install required libraries

```
pip install numpy pandas matplotlib seaborn scikit-learn jupyter
```

4. Start Jupyter Notebook

```
jupyter notebook
```

5. Open

```
credit_wise.ipynb
```

6. Run all cells

Make sure the dataset is available in the same folder as the notebook.

■ Key Learning Outcomes

Through this project, the following Machine Learning concepts were implemented:
• Data loading and inspection
• Missing value treatment
• Exploratory Data Analysis
• Data visualization
• Label Encoding
• One-Hot Encoding
• Correlation analysis
• Train-Test Split
• Feature Scaling
• Classification algorithms
• Model evaluation
• Feature engineering

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■ Project

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